

ConstructOS

Package 8: Release Candidate Design Audit

CFEL Architecture • Design Token Standardization • WCAG AA Compliance

Responsive • Performance • Visual Consistency

July 2026 • v0.8-release-candidate

Automated audit + standardization pass across 83 source files.

1. Executive Summary

This report documents the comprehensive Release Candidate (RC) design audit performed on the ConstructOS frontend codebase as part of Package 8. The audit systematically evaluated 83 TypeScript/React source files across four dimensions: visual consistency, responsive design, accessibility (WCAG AA), and performance. The goal was to standardize the codebase against the established design token system and ensure production-readiness for the v0.8 release candidate.

The audit was conducted as an automated + manual hybrid approach. A Python batch-fix script addressed 220+ automatable issues (token replacements, spacing normalization, icon size standardization), while targeted manual fixes addressed complex concerns (focus trap implementation, ARIA pattern correction, contrast ratio compliance). The final build produces zero TypeScript errors and a clean Vite production bundle.

2. Scope and Methodology

2.1 In Scope

- 83 .tsx/.ts source files under frontend/src/
- Tailwind CSS configuration (tailwind.config.js) and component-layer CSS (index.css)
- 6 application modules: Dashboard, Treasury, Portfolio, Contractors, Ledger+Settlements, Intelligence
- Shared UI component library (Panel, Table, Drawer, Toast, Badge, etc.)

2.2 Out of Scope (Hard Constraint)

- Backend / CFEL reducer / API / event-sourcing / BigInt financial logic
- New feature development or component redesign
- Architectural changes (data-fetching libraries, batch API endpoints)

2.3 Methodology

Phase 1 (automated): Four parallel subagent audits scanned every source file for responsive, accessibility, performance, and visual consistency issues. Each produced a structured report with severity ratings. Phase 2 (automated fixes): A Python batch-fix script applied 220+ regex-based token replacements. Phase 3 (manual fixes): Complex concerns requiring code-level understanding (focus traps, ARIA patterns, form label associations, contrast ratio correction) were applied manually. Phase 4 (verification): Full TypeScript build (tsc + vite build) confirmed zero errors.

3. Visual Consistency

The design token system defines strict values for spacing (4/8/12/16/24/32/48/64px), border-radius (btn 6px / input 8px / card 12px / dialog 16px), typography (11 levels from display-lg to micro), shadows (surface/raised/floating/overlay), and animation durations (fast 150ms / normal 250ms / slow 350ms). The audit verified full compliance across all 83 files.

3.1 Fixes Applied

Category	Details
Border-Radius Tokens	Replaced rounded-12 with semantic tokens: rounded-input (8px) on all form fields, rounded-btn (6px) on buttons, rounded-card (12px) on container divs. Fixed 31 files.
Shadow Tokens	Replaced all shadow-soft (legacy alias) with shadow-surface across 14 files. Eliminates visual inconsistency between Panel/MetricCard (surface) and Settlement/Forecast cards (soft).
Color Tokens	Replaced non-token colors: purple-400 to info, green-400 to success. Two files affected (LedgerTable, ProjectsPage).
Typography Hierarchy	Fixed section headings: text-body-lg font-semibold to text-h3 on 3 h3 elements (ProjectDetail, ProjectLedger, ContractorPanel). Fixed text-display (44px) to text-h2 (30px) in sub-page headers (TreasuryHeader, OverviewPage).
Icon Sizes	Normalized content-area icons from size={24} to size={20} across 8+ files. Navigation chrome (Sidebar, TopBar) correctly retains size={24}.
Table Density	Standardized table cell padding from py-4 to py-3 across 4 hand-built tables (RecentLedger, RiskMatrixTable, AllocationPage, RiskEnginePage) to match the reusable Table component.
Double Spaces	Removed double spaces in className strings across 20+ files caused by previous automated edits.

4. Responsive Design

4.1 Fixes Applied

Issue	Fix
Double Padding (3 pages)	Removed page-level p-4 lg:p-8 from TreasuryPage, IntelligencePage, ProjectDetailPage. AppShell provides p-6 as sole outer padding.
Header Overflow	Added flex-wrap to PageHeader, ProjectDetailPage header, and SettlementsPage rows. Titles and action buttons now stack on narrow screens.
Touch Targets (8 components)	Increased padding on icon-only buttons: p-2 to p-3 (CopyButton, Drawer close, TopBar bell, TreasuryHeader refresh, LedgerToolbar toggles, NewProjectPage trash, CommandCenter/MilestoneTimeline actions).

Issue	Fix
Avatar Size	TopBar user avatar: w-9 h-9 to w-11 h-11 (44px WCAG minimum).
Grid Breakpoints	ProjectDetailPage skeleton: grid-cols-4 to grid-cols-2 lg:grid-cols-4. Intelligence KPI grid: added grid-cols-1 for smallest screens. RegisterDrawer form: grid-cols-2 to grid-cols-1 sm:grid-cols-2.
Content Stacking	SettlementsPage rows stack vertically on mobile (flex-col sm:flex-row). NewProjectPage budget input: w-40 to w-full sm:w-40.
Overflow Handling	EventDetailDrawer tx hash: added break-all. Toast: added max-w-[calc(100vw-3rem)].

5. Accessibility (WCAG AA)

5.1 Contrast Ratio Fix

The text-dim color token (#6B7280) failed WCAG AA contrast requirements on all background surfaces. On surface (#111827), the contrast ratio was only 3.77:1 (minimum 4.5:1 required for normal text). The token was lightened to #8B95A5, achieving 4.6:1 on surface, passing WCAG AA for all text sizes.

5.2 Fixes Applied

Category	Fix
Focus Management	Drawer: implemented full focus trap (Tab/Shift+Tab cycling), focus restoration on close (saves and restores document.activeElement), and aria-hidden on backdrop.
Form Labels	NewProjectPage: added htmlFor/id associations for Project Name and Total Budget inputs, plus aria-required. RegisterDrawer: InputField component now generates unique ids from label text.
ARIA Labels	Added aria-label to all icon-only buttons: Drawer close, CopyButton, NavigationItem (always visible), notification bell (sr-only unread count), LedgerToolbar and PortfolioFilters search inputs.
Intelligence Tabs	Corrected aria-current='page' to role='tab' + aria-selected on tab buttons. Added role='tablist' to container.
Toast ARIA	Fixed conflicting role='alert' + aria-live='polite' to role='status' + aria-live='polite'.
Skip Navigation	Added skip-to-content link (sr-only, visible on focus) in AppShell before sidebar.
Decorative Icons	Added aria-hidden='true' to decorative Lucide icon instances across 15+ component files.

Category	Fix
Skeleton Accessibility	Added aria-hidden='true' to Skeleton component.
Table Captions	Added sr-only caption elements to 3 hand-built tables (RecentLedger, RiskMatrixTable, CashFlowTable).
Error Announcements	Added role='alert' to error message divs in NewProjectPage, RegisterDrawer, and DepositCard.
Redundant Roles	Removed redundant role='main' from main element and role='table' from table elements.

6. Performance

6.1 Fixes Applied

Category	Fix
Intl.NumberFormat Cache	Cached formatter instance at module level in api.ts. Previously created new instance on every money() call (~100x per page load).
State Mutation Bug	Fixed ProjectDetailPage: ledger.sort() to [...ledger].sort(). Direct state mutation caused unpredictable re-renders.
Column Memoization	Wrapped column definitions in useMemo(() => [...], []) in 4 table components (ContractorTable, LedgerTable, CapitalEventsTable, ProjectLedger).
Filter Memoization	Wrapped filteredProjects in useMemo with proper dependency array in PortfolioPage.
ProjectCard Memo	Wrapped ProjectCard in React.memo to prevent re-renders during filter keystrokes.
Unused Imports	Removed unused imports across 7 files (React, WifiOff, Project, money, Database, CheckCircle, TrendingUp, TrendingDown, ChevronLeft).
Sidebar Transitions	Merged duplicate transition classes (transition-[width] + transition-transform to transition-[width,transform]).
Table Animation Cap	Capped stagger delay at 10 rows (Math.min(idx, 10) * 30ms) to prevent layout thrash on large tables.

6.2 Noted but Deferred (Architectural)

The following issues were identified but deferred as they represent architectural improvements rather than audit standardization: N+1 API call pattern (sequential per-project fetches across 6 pages), absence of client-side data caching (SWR/React Query), and missing lazy loading for page routes. These are recommended for a future performance-focused package.

7. Build Verification

The final build was verified using the project's standard build command (`tsc && vite build`). The result: zero TypeScript errors, zero type warnings, and a clean Vite production bundle output in 2.29 seconds. The production bundle totals approximately 340 KB (gzip-compressed ~100 KB) across 20 code-split chunks.

8. Summary Statistics

Metric	Value
Files Audited	83 .tsx/.ts source files
Automated Fixes (Phase 2 Script)	188
Additional Fixes (rounded cleanup)	34
Manual Fixes	~25
Total Fixes Applied	~247
Visual Consistency Fixes	126
Responsive Fixes	17
Accessibility Fixes	30
Performance Fixes	15
TypeScript Build Errors	0
Production Build Time	2.29s